Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.30768561341733486, median 0.24704463747835764, std: 0.22986441021971657
Gyroscope error (imu0): mean 0.1469930165686941, median 0.12060129730327536, std: 0.10254524734073675
Accelerometer error (imu0): mean 0.09576159087574936, median 0.07358415835144828, std: 0.09098871119245694

Residuals

Reprojection error (cam0) [px]: mean 0.30768561341733486, median 0.24704463747835764, std: 0.22986441021971657
Gyroscope error (imu0) [rad/s]: mean 0.004567672537554537, median 0.003747574181037265, std: 0.0031864990668868066
Accelerometer error (imu0) [m/s^2]: mean 0.06004005789552572, median 0.04613537731789395, std: 0.057047584922879806

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99893785 -0.04603703 -0.00193816 0.00180358]
 [0.00522068 0.15487279 -0.98792063 -0.00079501]
 [0.0457811   0.98686119 0.15494864 -0.03285867]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.99893785 0.00522068 0.0457811 -0.00029321]
 [-0.04603703 0.15487279 0.98686119 0.0326331]
 [-0.00193816 -0.98792063 0.15494864 0.00430949]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.009774567651316993

Gravity vector in target coords: [m/s^2]
[0.0000504 -9.80483596 -0.18334299]

Calibration configuration
==========

cam0

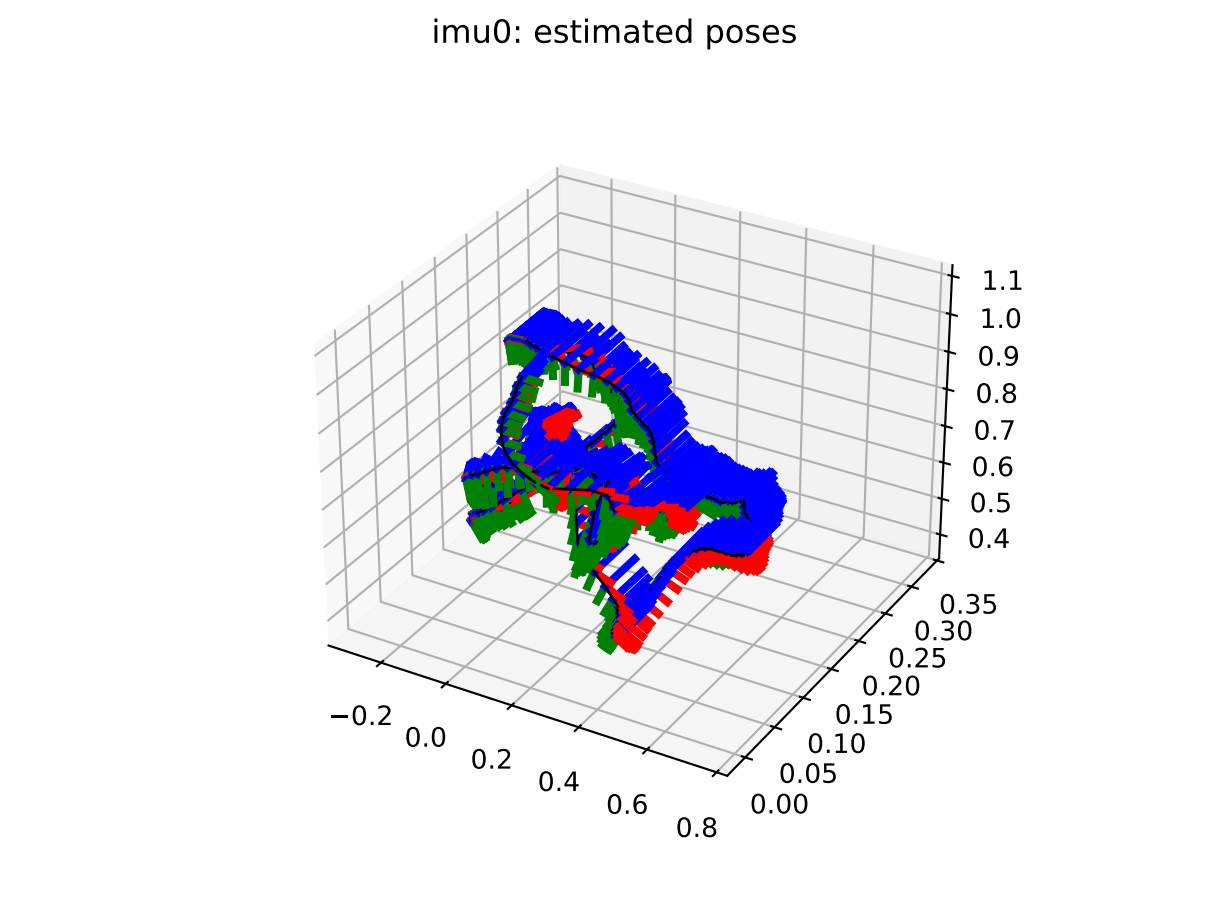
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

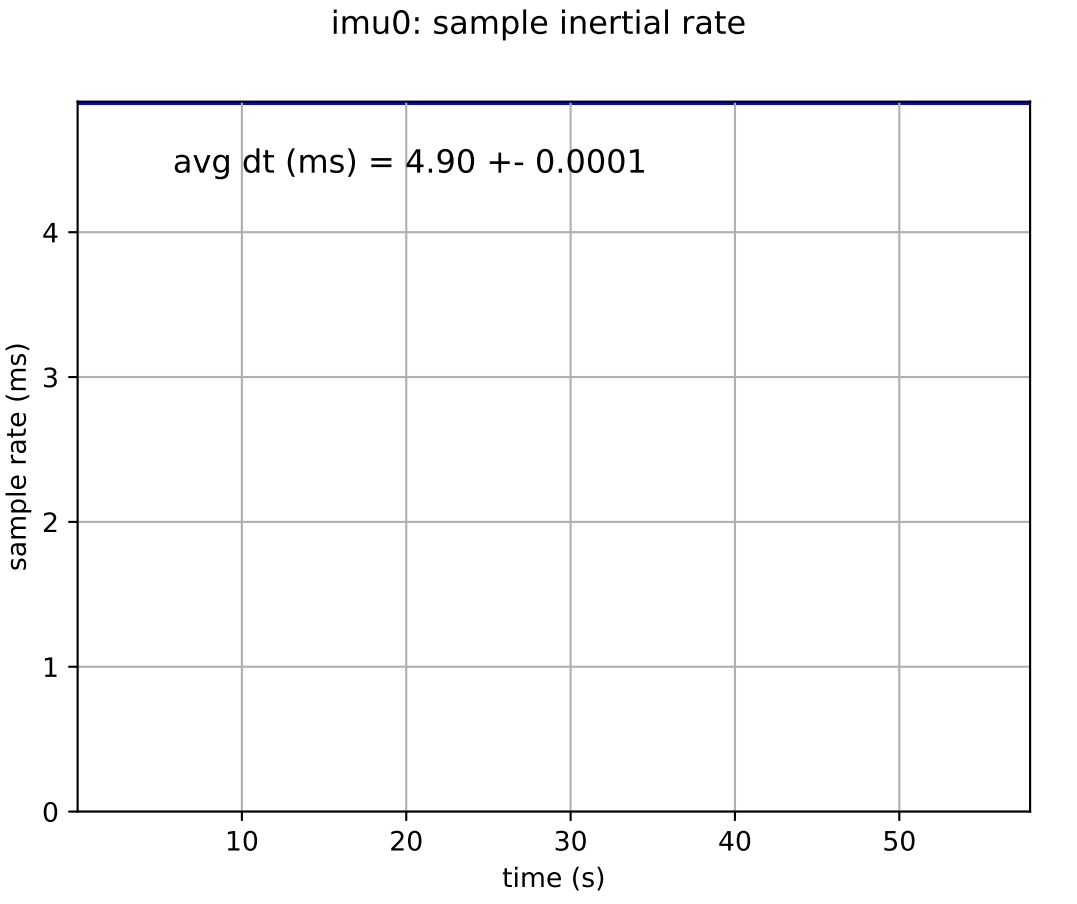
IMU configuration
==========

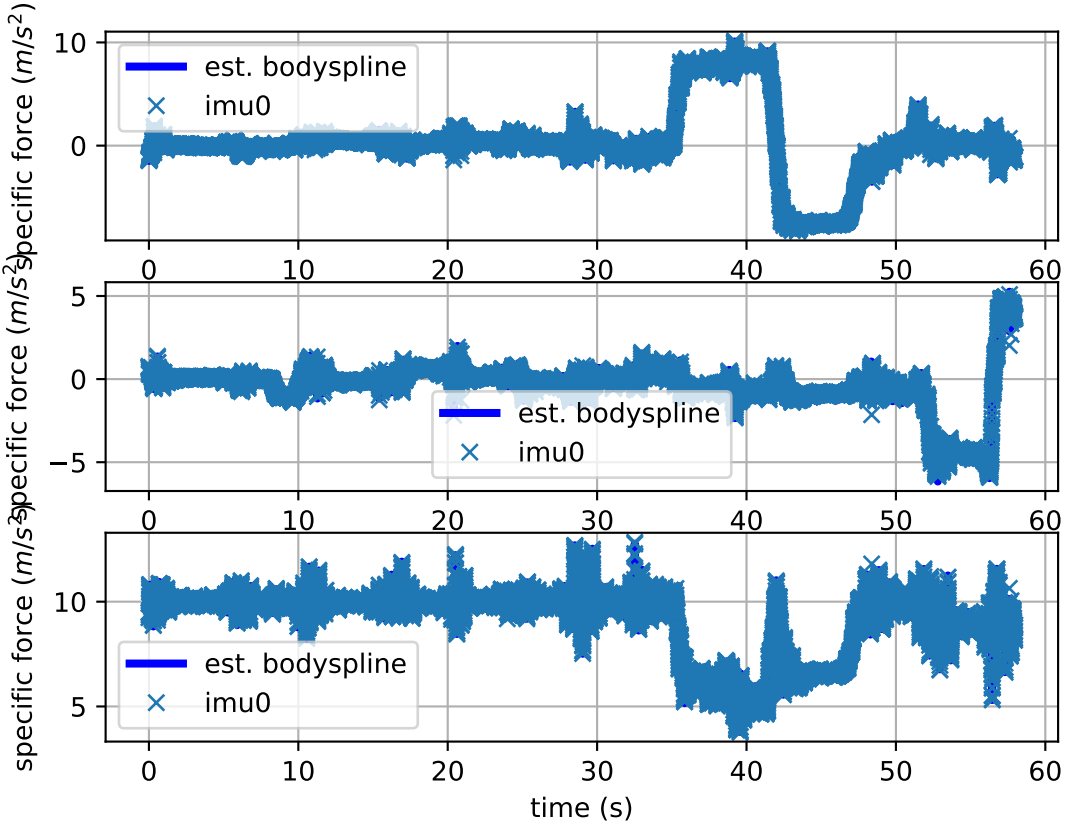
IMU0:

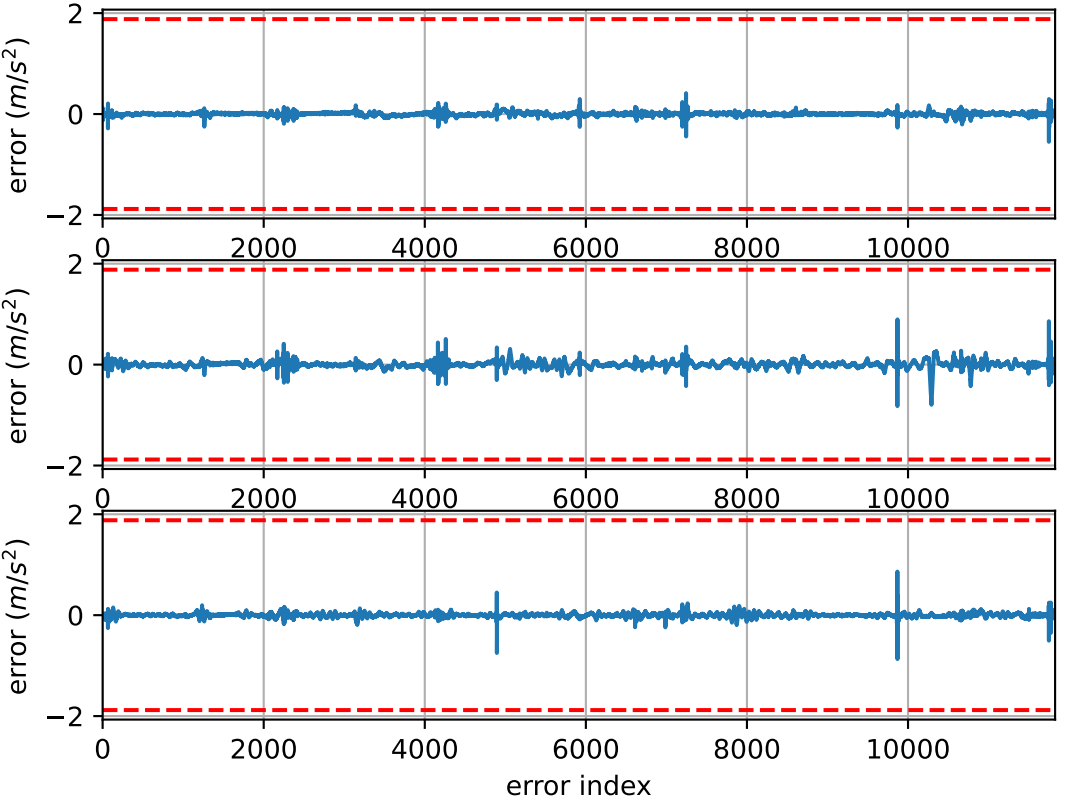
  Model: calibrated
  Update rate: 204.0355927763645
  Accelerometer:
    Noise density: 0.043893153414587355
    Noise density (discrete): 0.6269743155523354
    Random walk: 0.026159638330733644
  Gyroscope:
    Noise density: 0.002175430893269639
    Noise density (discrete): 0.03107407851188584
    Random walk: 0.0002234548981032961

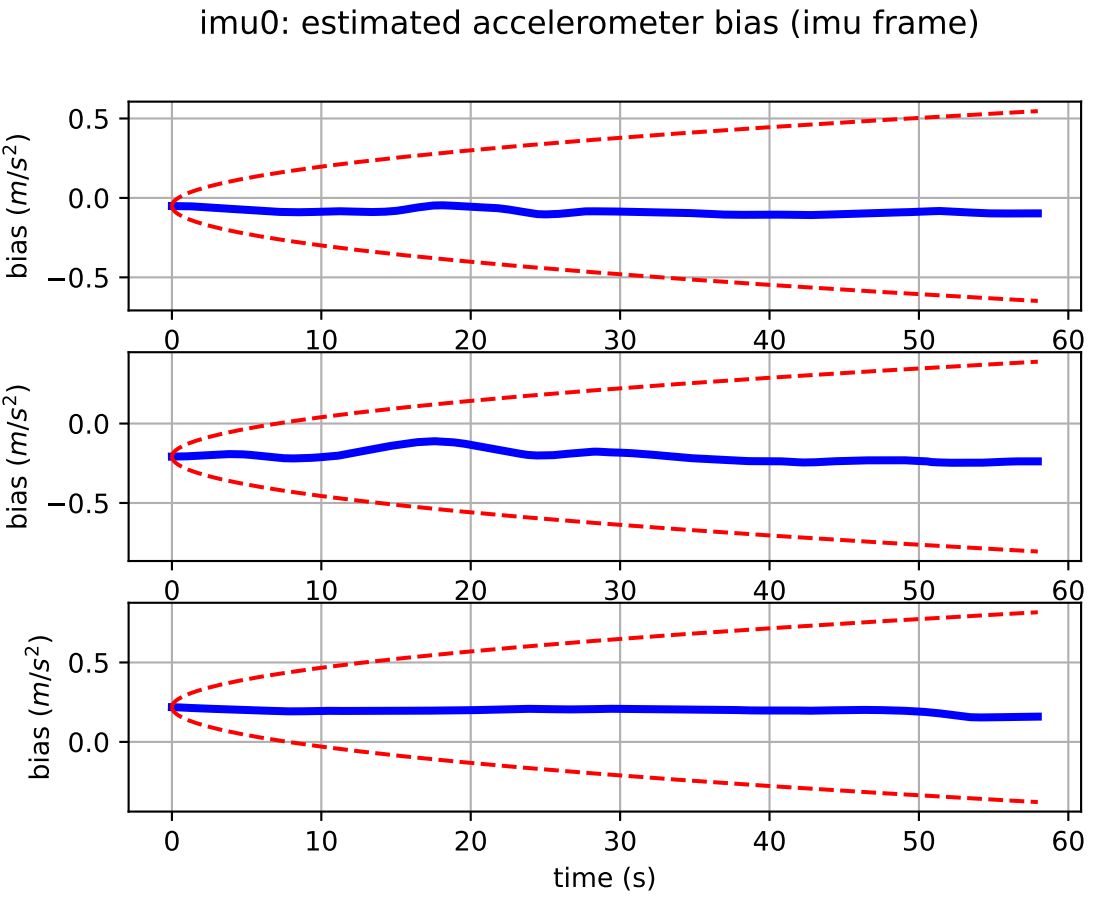
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

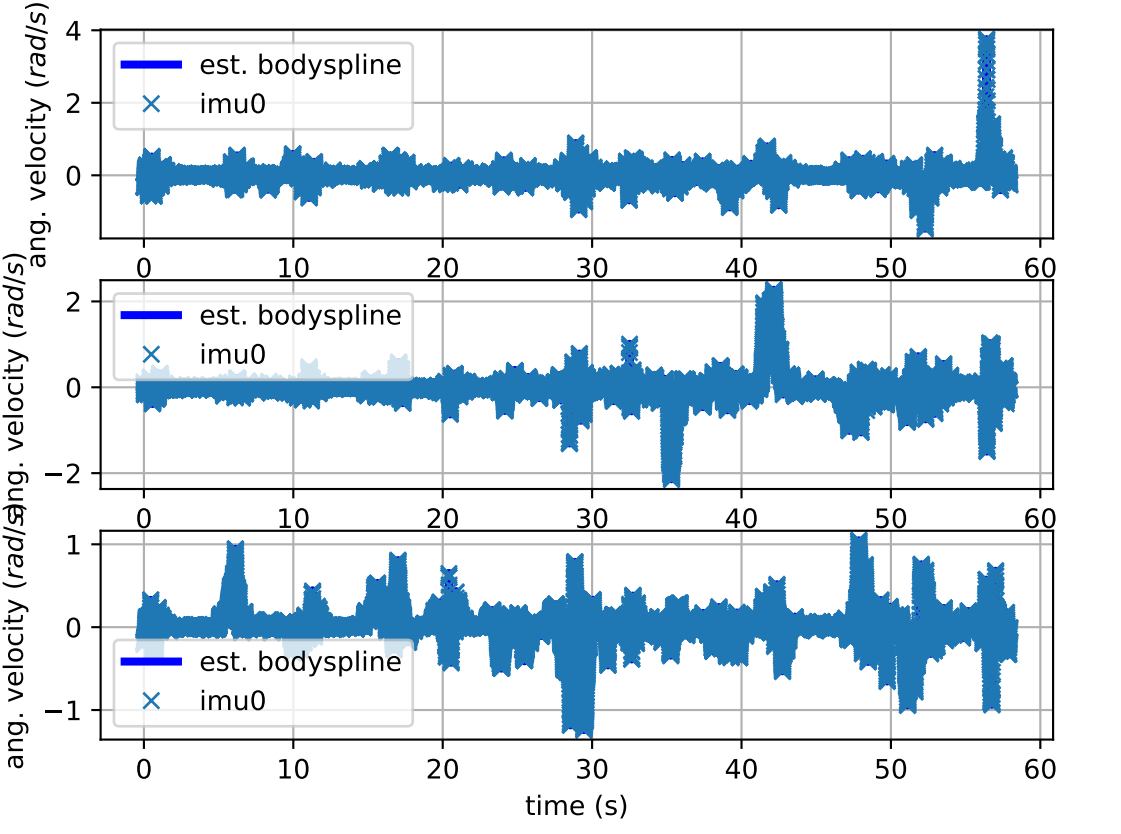

imu0: estimated poses
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
0.0
0.2
0.4
0.6
0.8

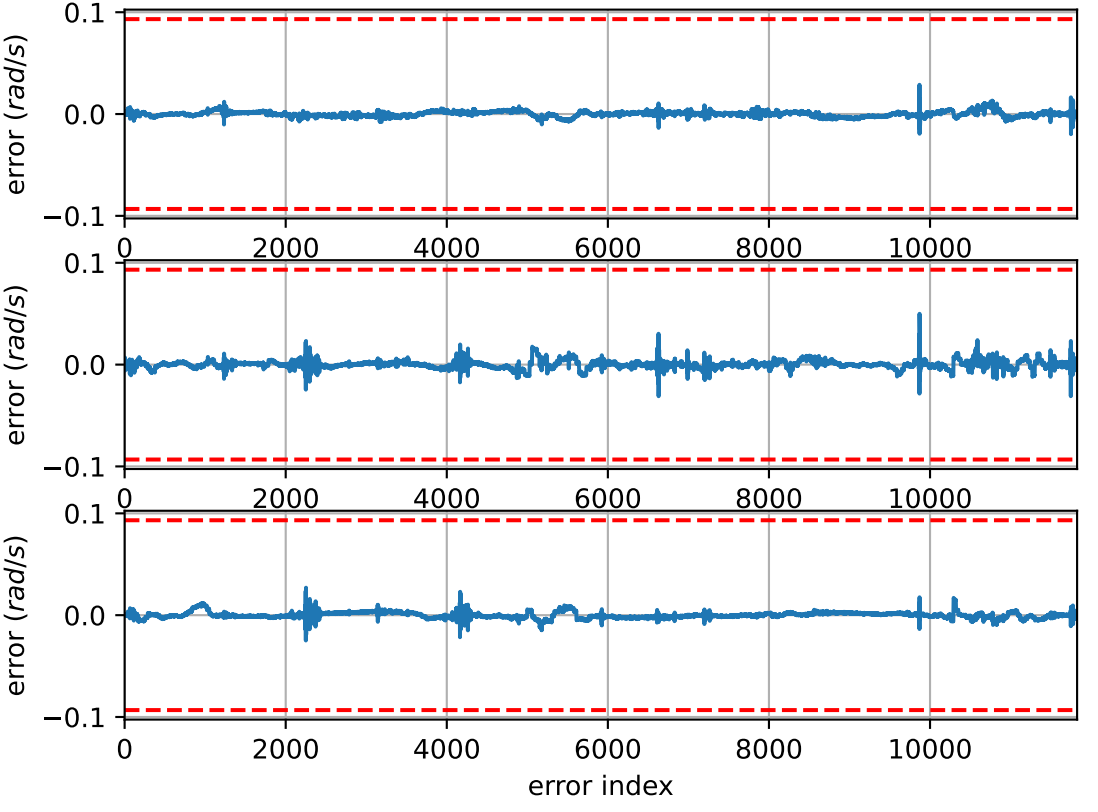

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0001
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

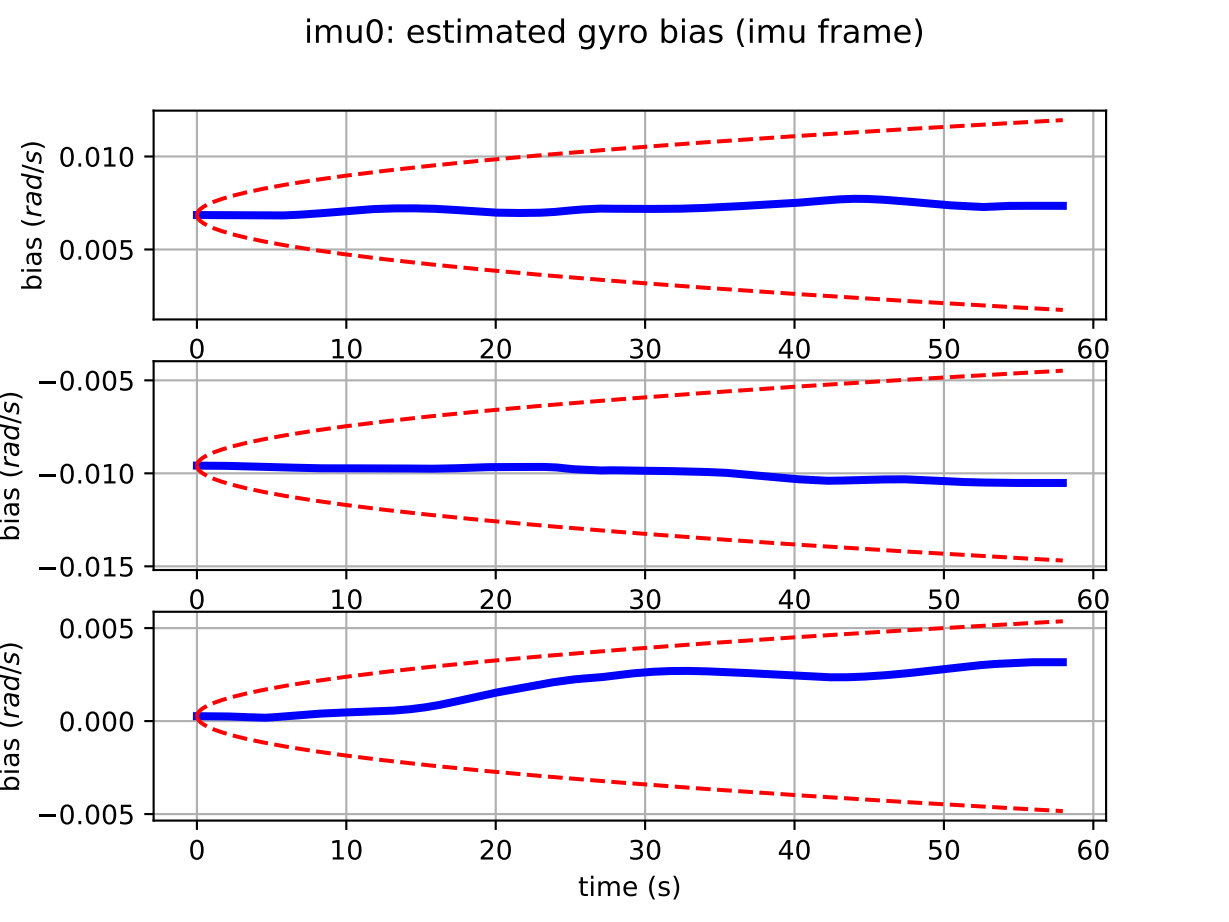

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

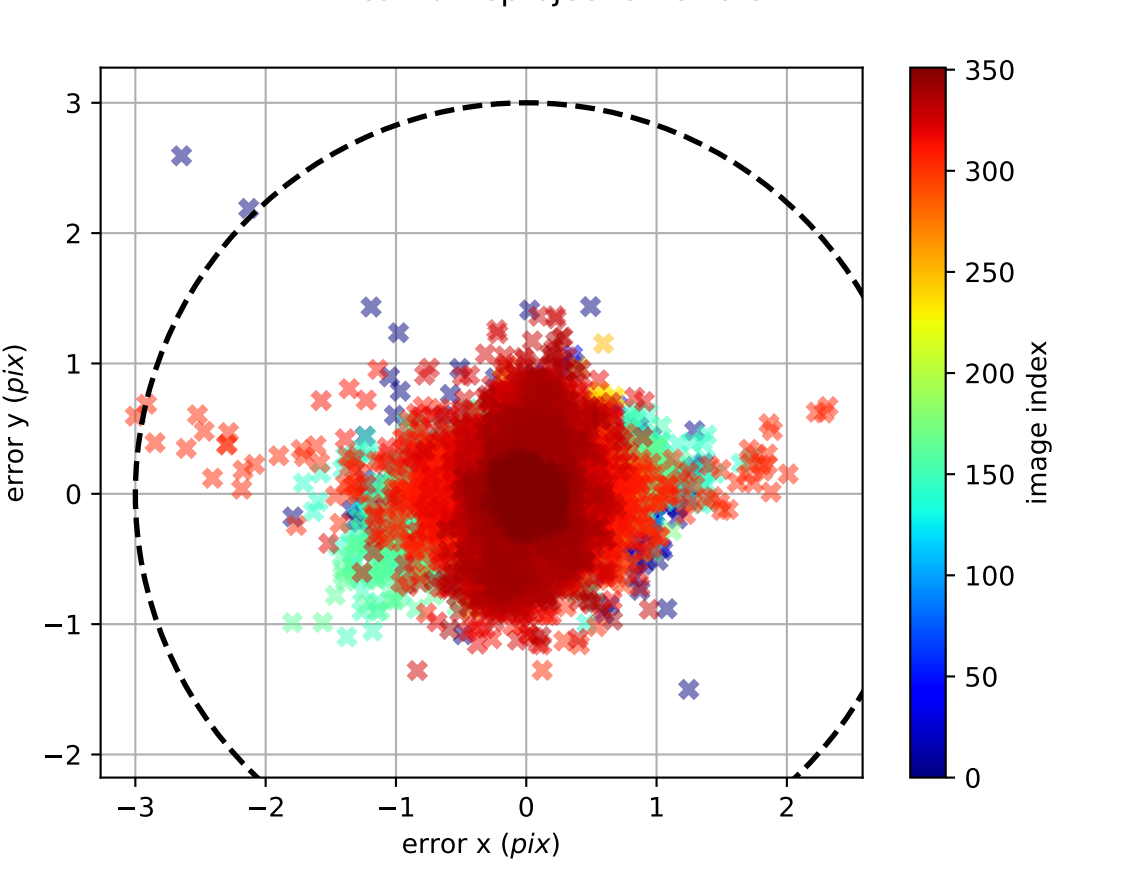

imu0: acceleration error
error (m/s^2)
2
0
−2
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.5
0.0
−0.5
0
10
20
30
40
50
60
bias (m/s^2)
0.0
−0.5
0
10
20
30
40
50
60
bias (m/s^2)
0.5
0.0
0
10
20
30
40
50
60
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
4
2
0
0
10
20
30
40
50
60
ang. velocity (rad/s)
est. bodyspline
imu0
2
0
−2
0
10
20
30
40
50
60
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
−1
0
10
20
30
40
50
60
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.005
0
10
20
30
40
50
60
bias (rad/s)
−0.005
−0.010
−0.015
bias (rad/s)
0.005
0.000
−0.005
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−3
−2
−1
0
1
2
3
0
50
100
150
200
250
300
350